TurtleBot3 SBC Setup

PINKLAB

Contents

TurtleBot3 SBC Setup

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SBC에 Ubuntu24.04 환경 구축하기

Raspberry Pi Imager 설치

Code

```
sudo apt update
sudo apt install rpi-imager
```

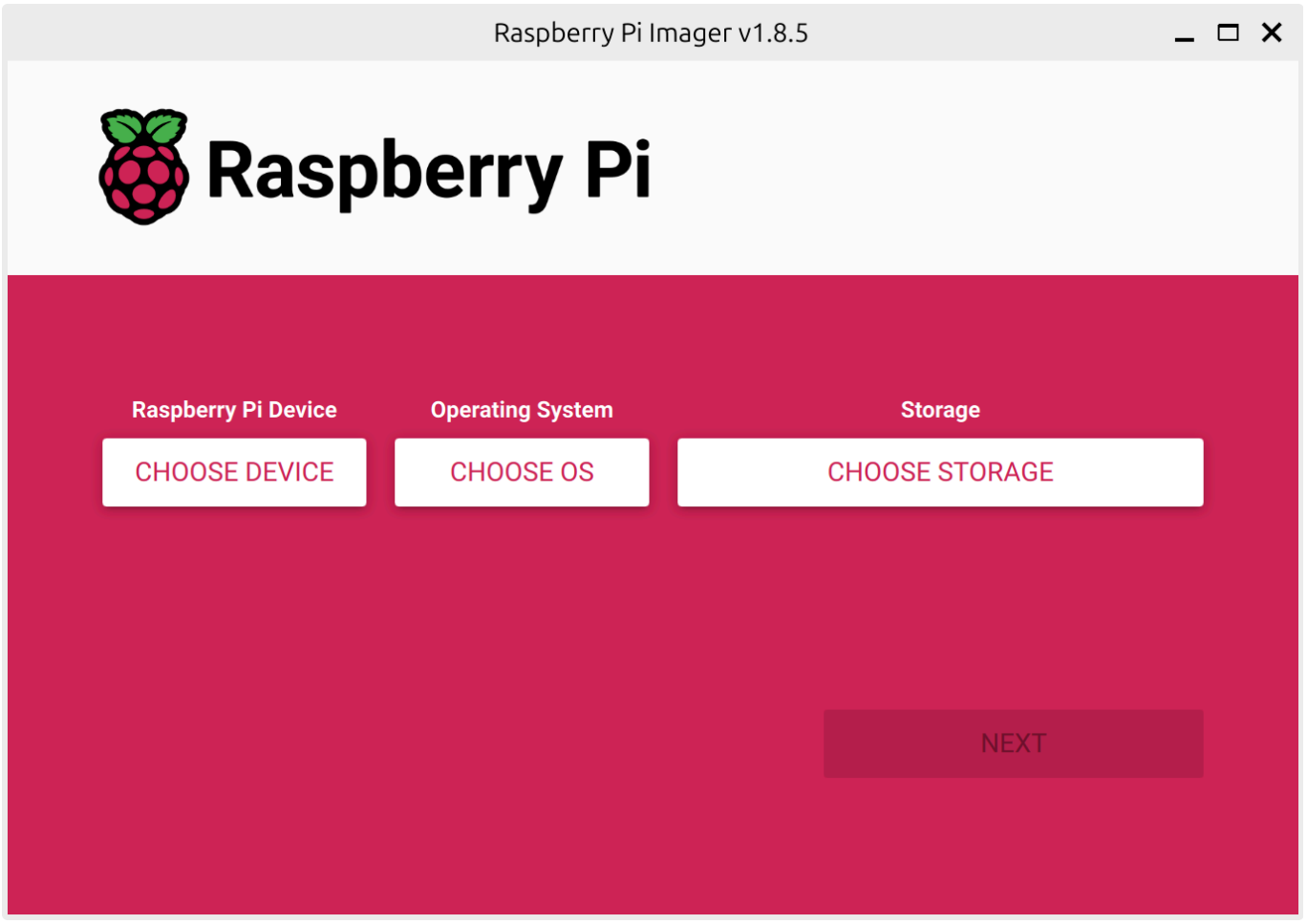
```
mkh@mkh:~$ sudo apt install rpi-imager
[sudo: authenticate] Password:
Installing:
  rpi-imager

Installing dependencies:
  javascript-common        libqt5waylandclient5
  libdouble-conversion3    libqt5waylandcompositor5
```

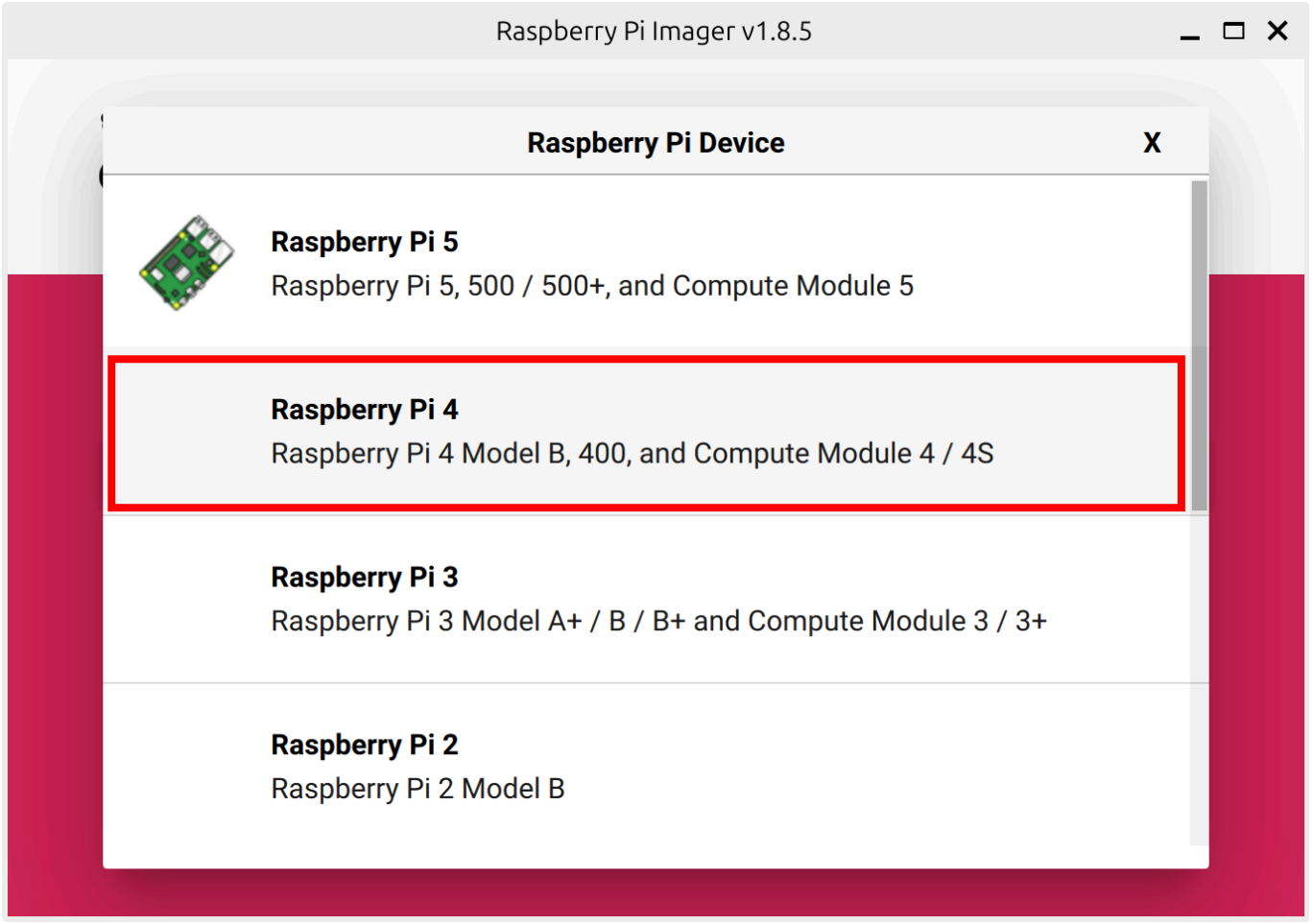
Raspberry Pi Imager 실행

Code

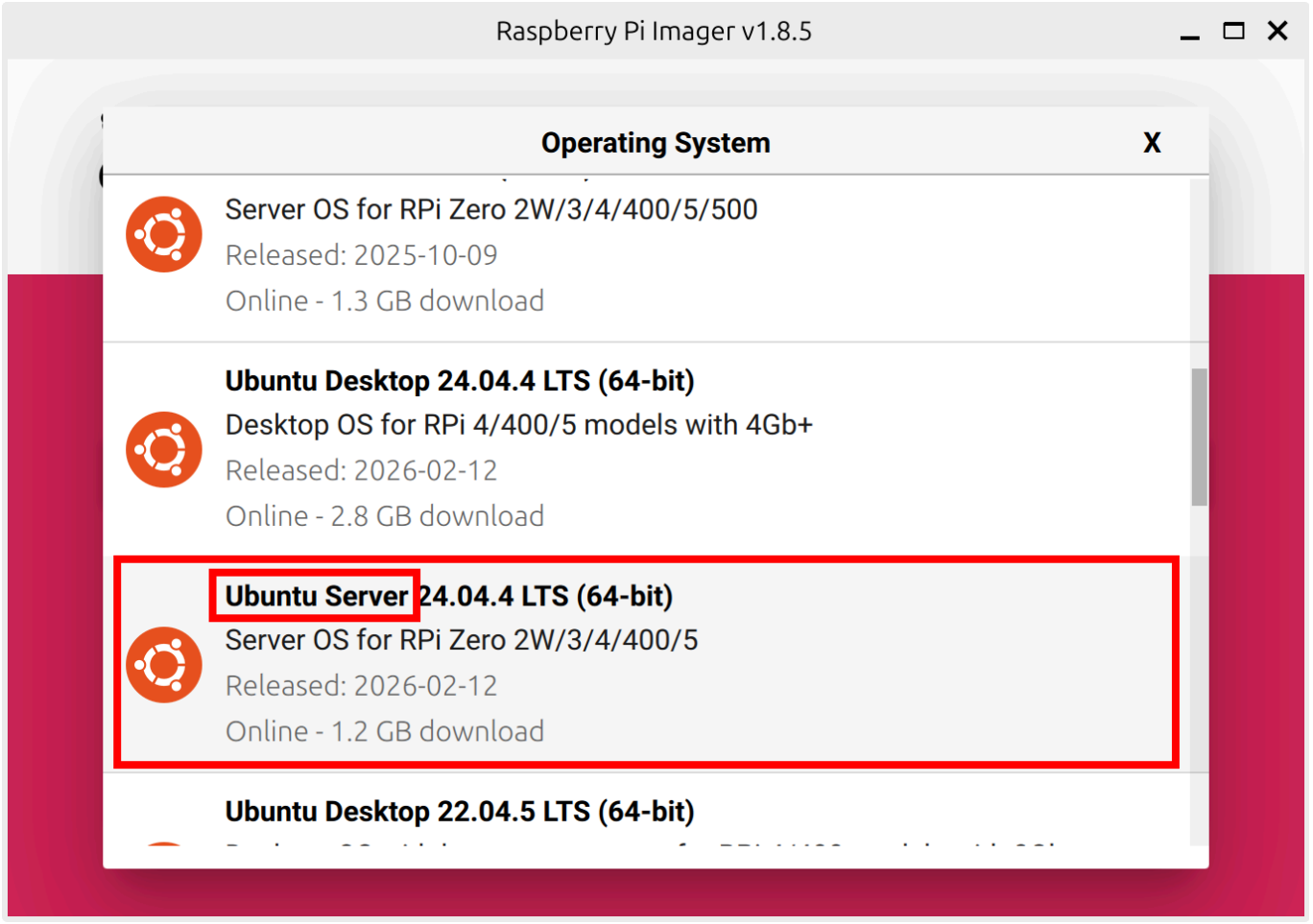
```
rpi-imager
```



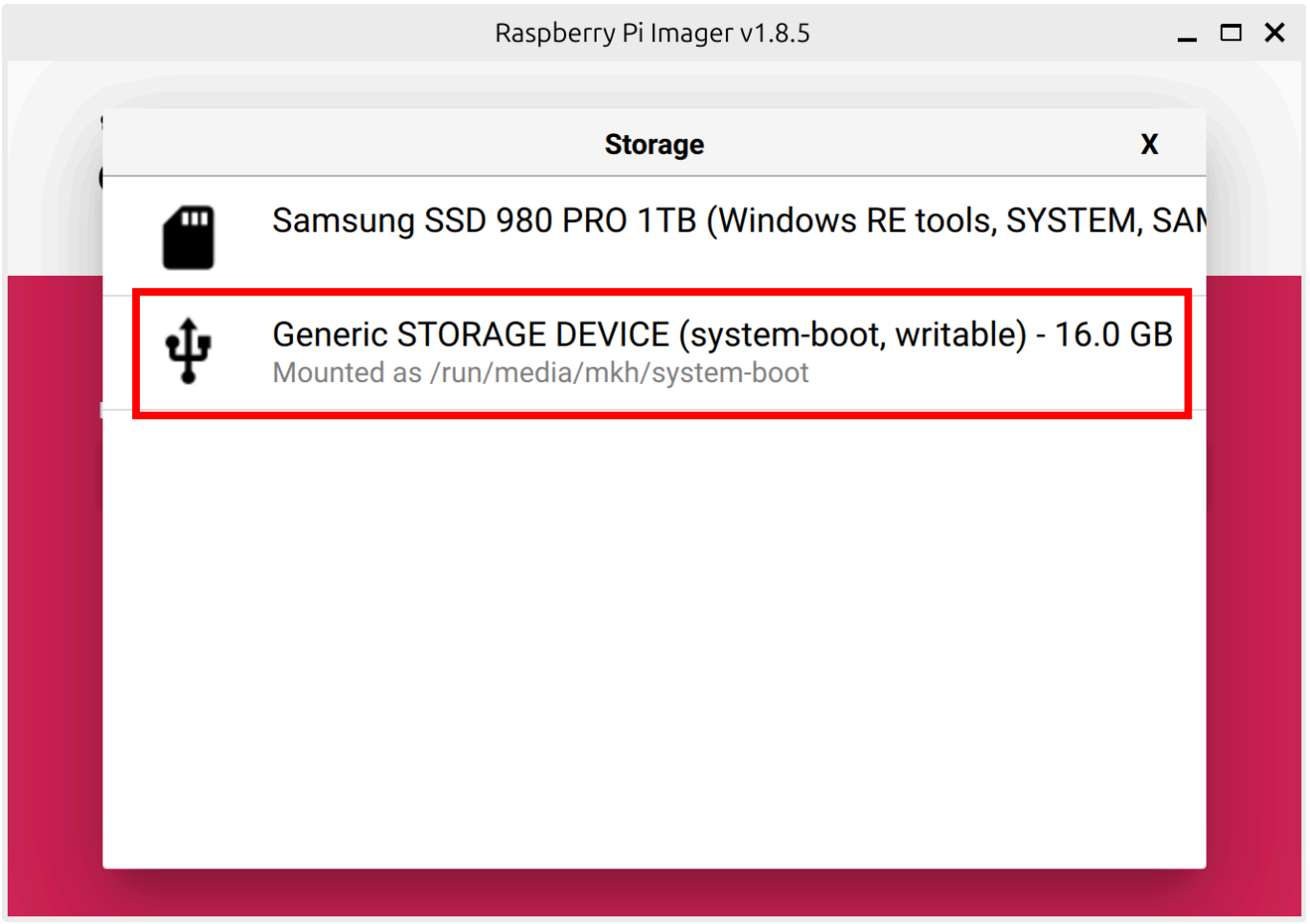
Raspberry Pi Device → Raspberry Pi 4 선택



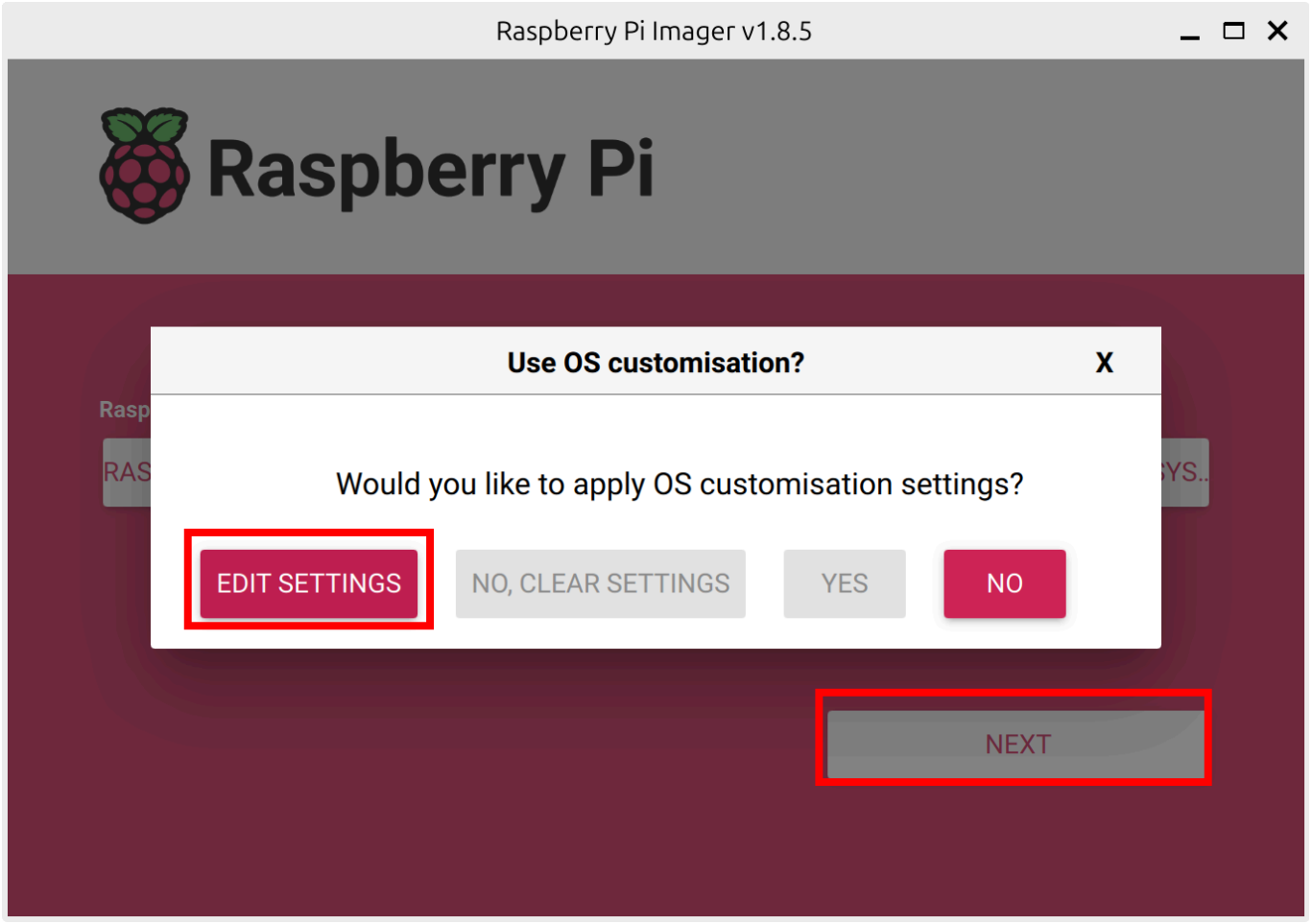
CHOOSE OS→ Other general-purpose OS → Ubuntu Ubuntu Server (64-bit) 선택



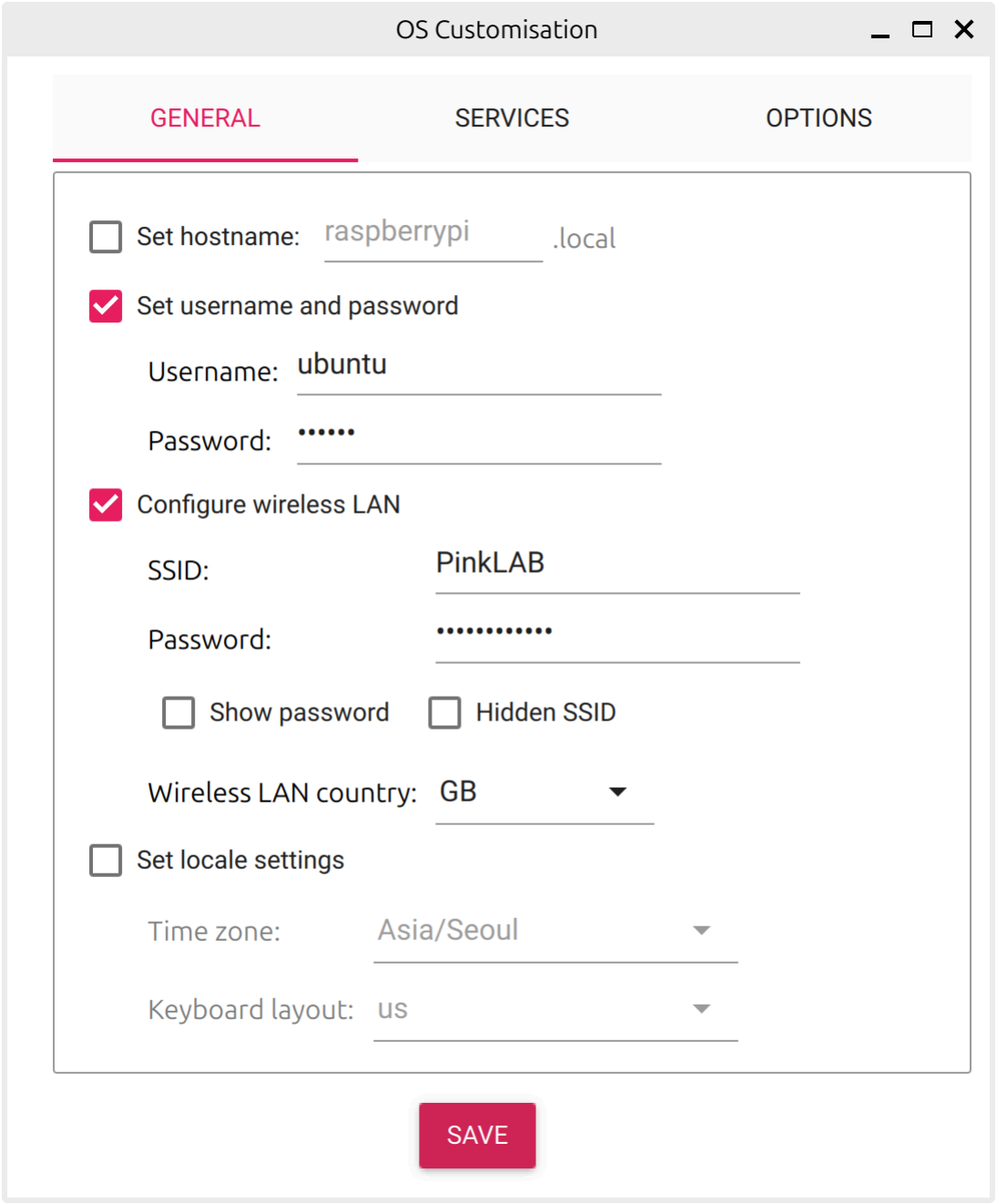
CHOOSE STORAGE에서 삽입한 microSD 카드 선택



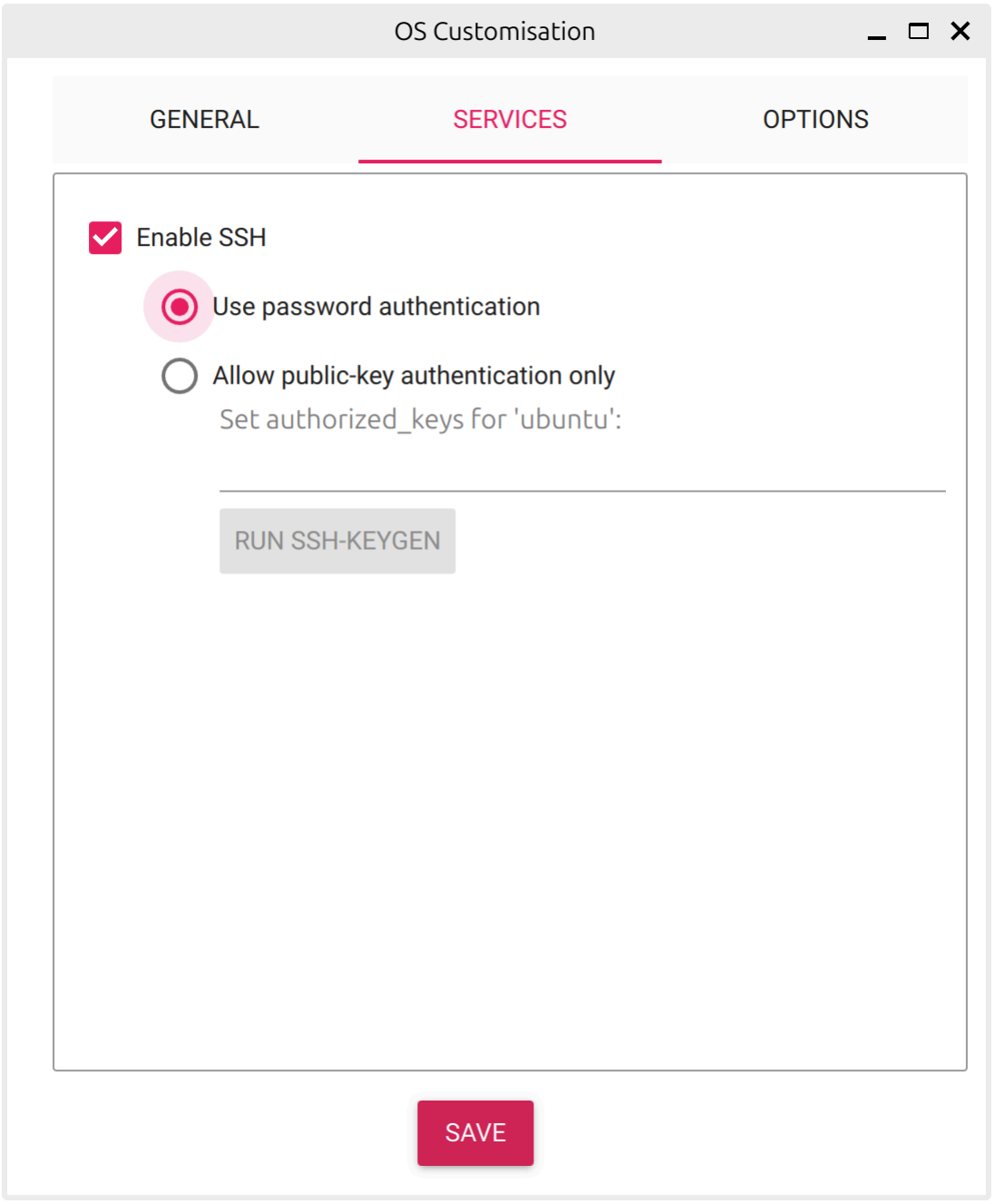
NEXT 클릭 후 EDIT SETTINGS 클릭



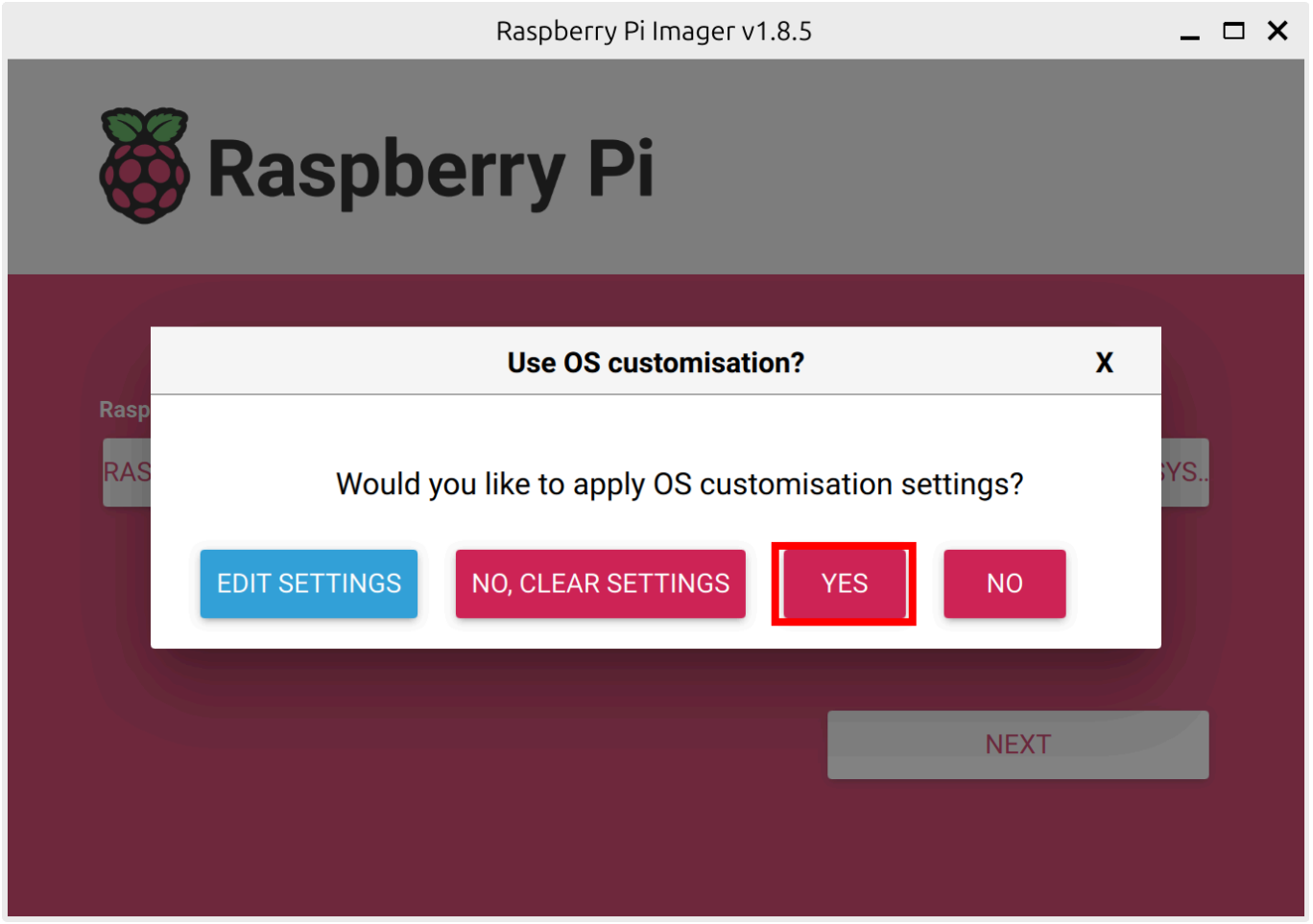
Username, Password, Wi-Fi 설정



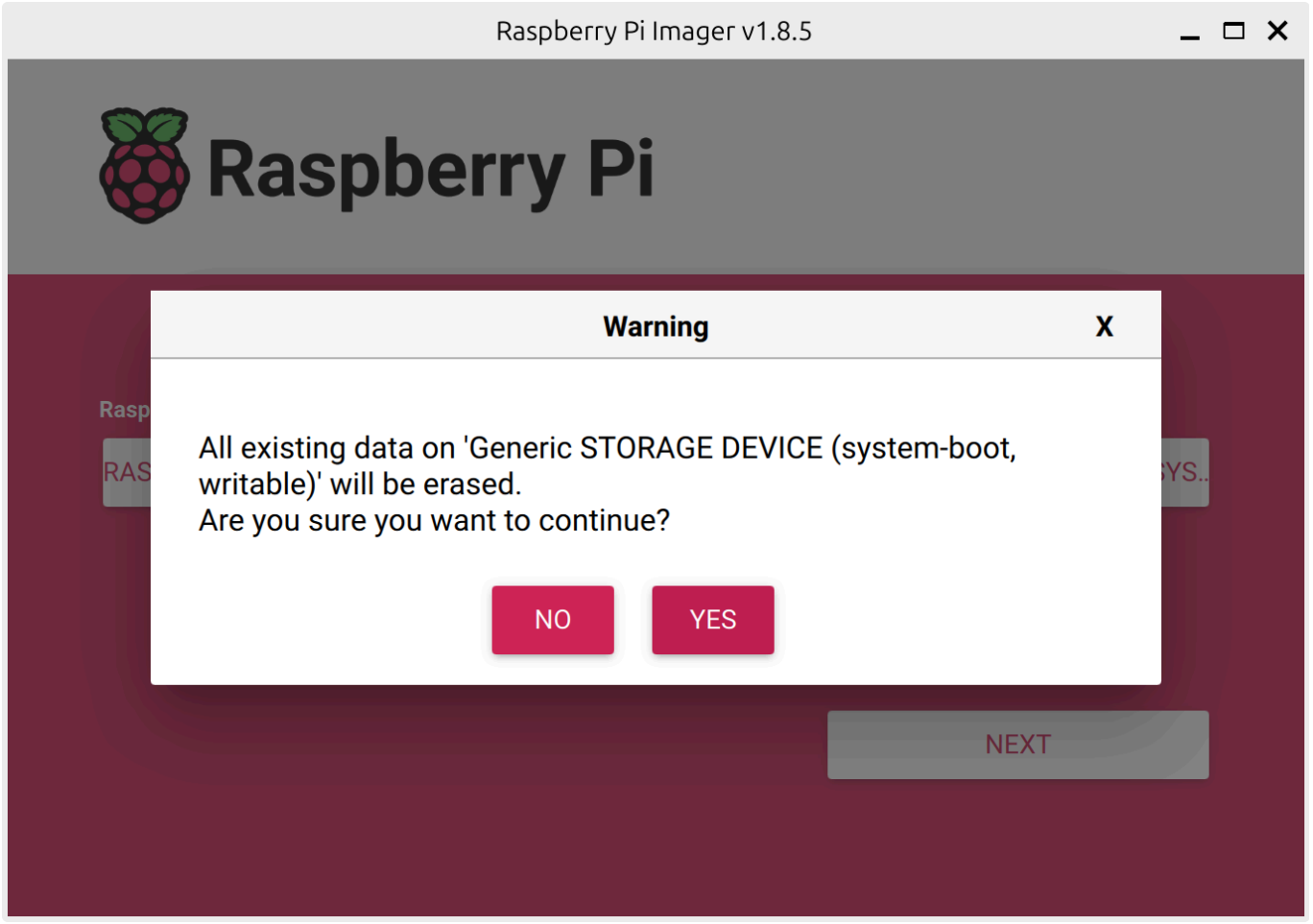
SERVICES 탭에서 SSH 활성화 후 SAVE


OS Customisation
GENERAL
SERVICES
OPTIONS
Enable SSH
Use password authentication
Allow public-key authentication only
Set authorized_keys for 'ubuntu':
RUN SSH-KEYGEN
SAVE

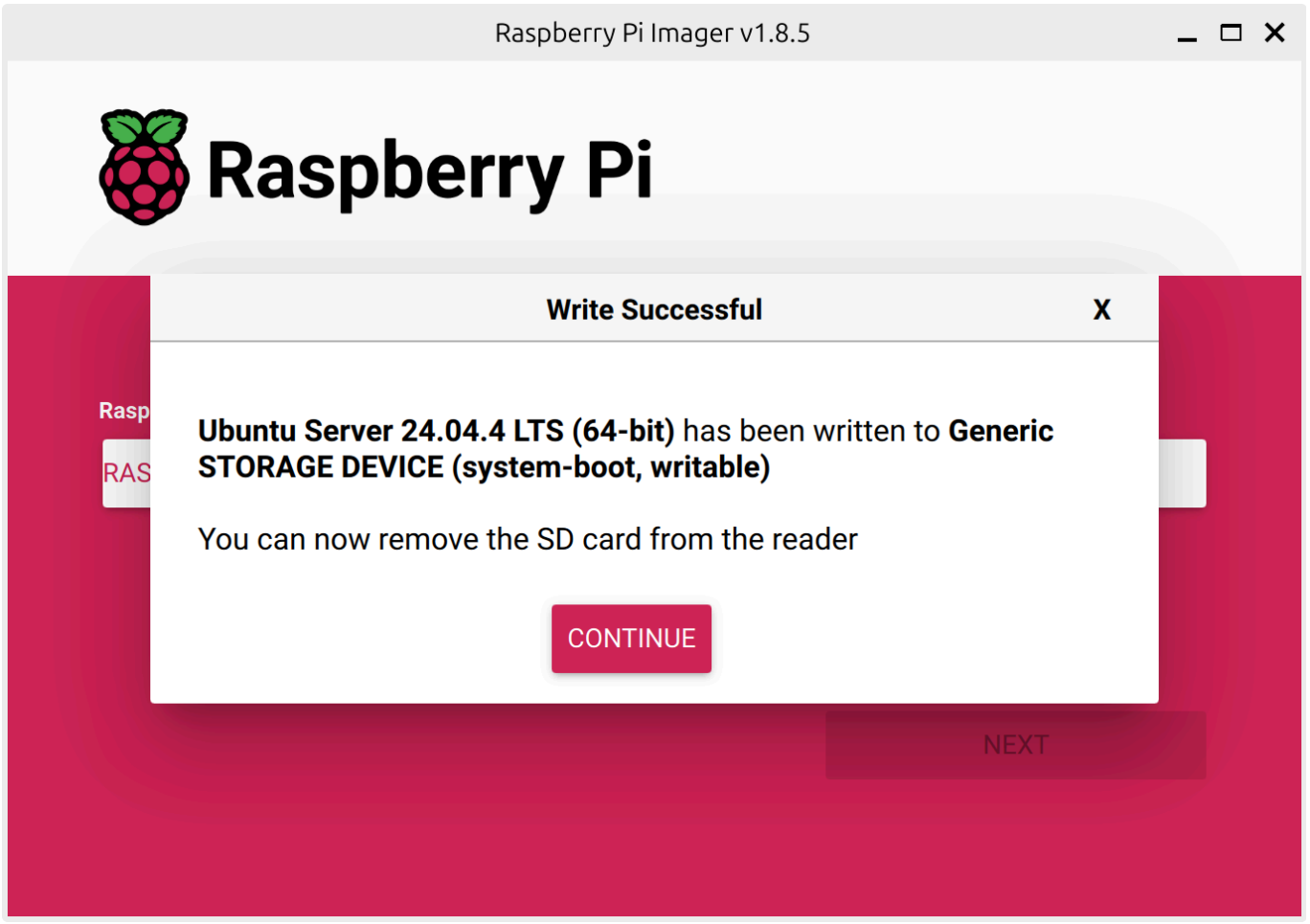
YES



YES 클릭 후 PC 비밀번호 입력 후 sd 카드 굽기 시작

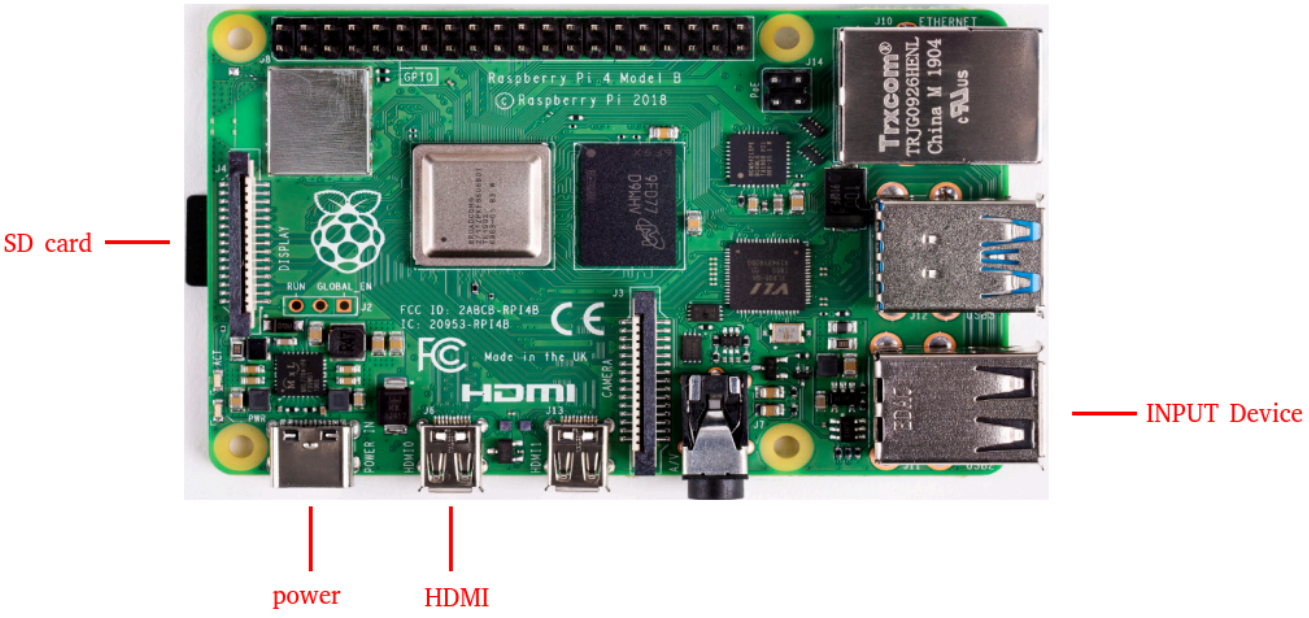

Raspberry Pi Imager v1.8.5
Raspberry Pi
Warning
X
All existing data on 'Generic STORAGE DEVICE (system-boot, writable)' will be erased.
Are you sure you want to continue?
NO
YES
NEXT

완료 되었다면



라즈베리파이에 microSD 삽입

HDMI, 키보드, 마우스 연결



배터리 연결하고, OpenCR과 라즈베리파이 전원 연결 확인

[SBC] 로봇이 할당받은 IP 주소 확인하기

Code

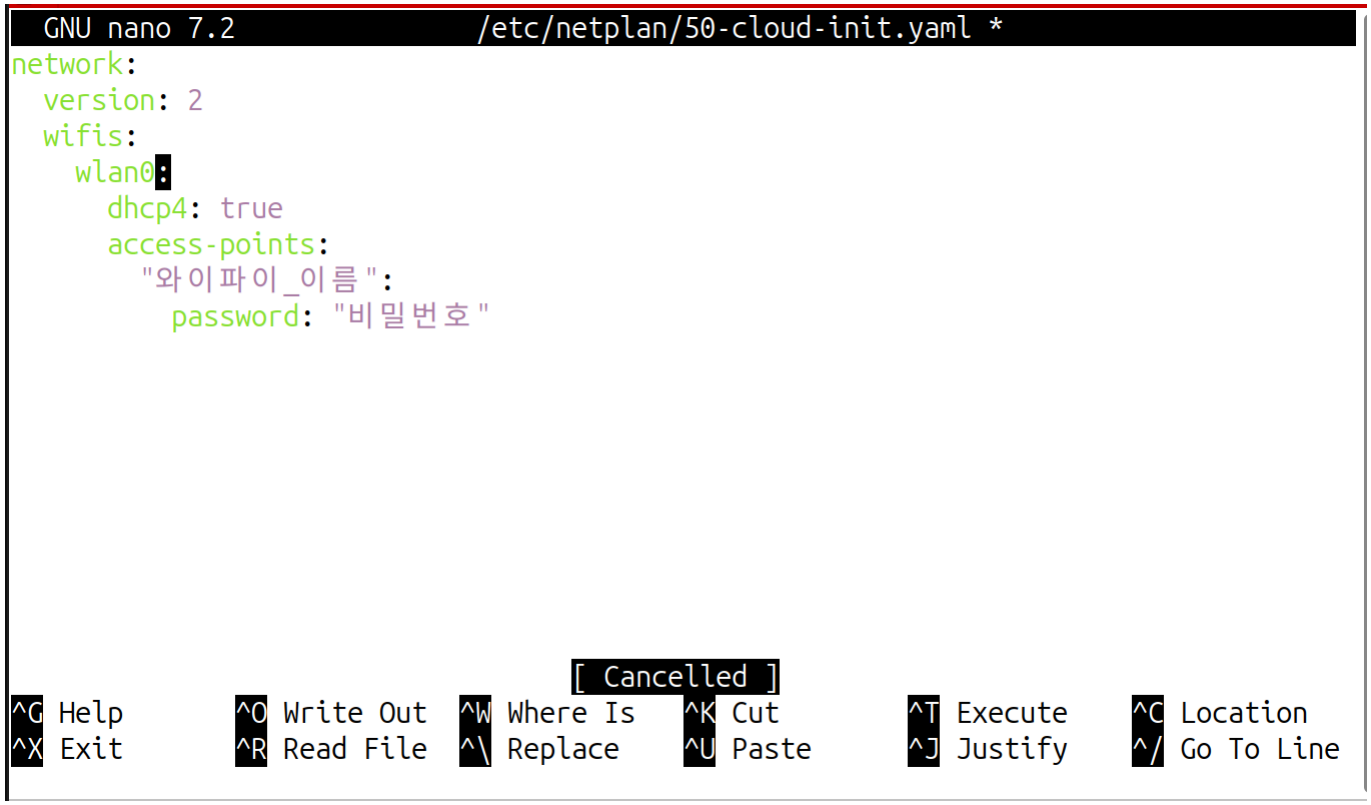
```
ip a
```

```
ubuntu@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default ql
en 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 100
0
    link/ether d8:3a:dd:22:dc:fb brd ff:ff:ff:ff:ff:ff
3: wlan0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group d
efault qlen 1000
    link/ether d8:3a:dd:22:dc:fc brd ff:ff:ff:ff:ff:ff
    inet 192.168.0.12/24 metric 600 brd 192.168.0.255 scope global dynamic wlan0
       valid_lft 7186sec preferred_lft 7186sec
    inet6 fe80::da3a:ddff:fe22:dcfc/64 scope link
       valid_lft forever preferred_lft forever
ubuntu@ubuntu:~$
```

[SBC] 만약에 공유기와 연결이 안 되어서 IP 주소가 안뜨는 경우 설정파일 열어서 적용

Code

```
sudo nano /etc/netplan/50-cloud-init.yaml
```



[SBC] 변경 적용 후 다시 확인하기

Code

```
sudo netplan apply
```

```
ubuntu@ubuntu:~$ sudo nano /etc/netplan/50-cloud-init.yaml
ubuntu@ubuntu:~$ sudo netplan apply
ubuntu@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default ql
en 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 100
0
    link/ether d8:3a:dd:22:dc:fb brd ff:ff:ff:ff:ff:ff
3: wlan0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group d
efault qlen 1000
    link/ether d8:3a:dd:22:dc:fc brd ff:ff:ff:ff:ff:ff
    inet 192.168.0.12/24 metric 600 brd 192.168.0.255 scope global dynamic wlan0
       valid_lft 7183sec preferred_lft 7183sec
    inet6 fe80::da3a:ddff:fe22:dcfc/64 scope link
       valid_lft forever preferred_lft forever
ubuntu@ubuntu:~$
```

[PC] PC에서 로봇으로 원격 접속하기

Code

```
ssh <username>@<로봇의 ip>
```

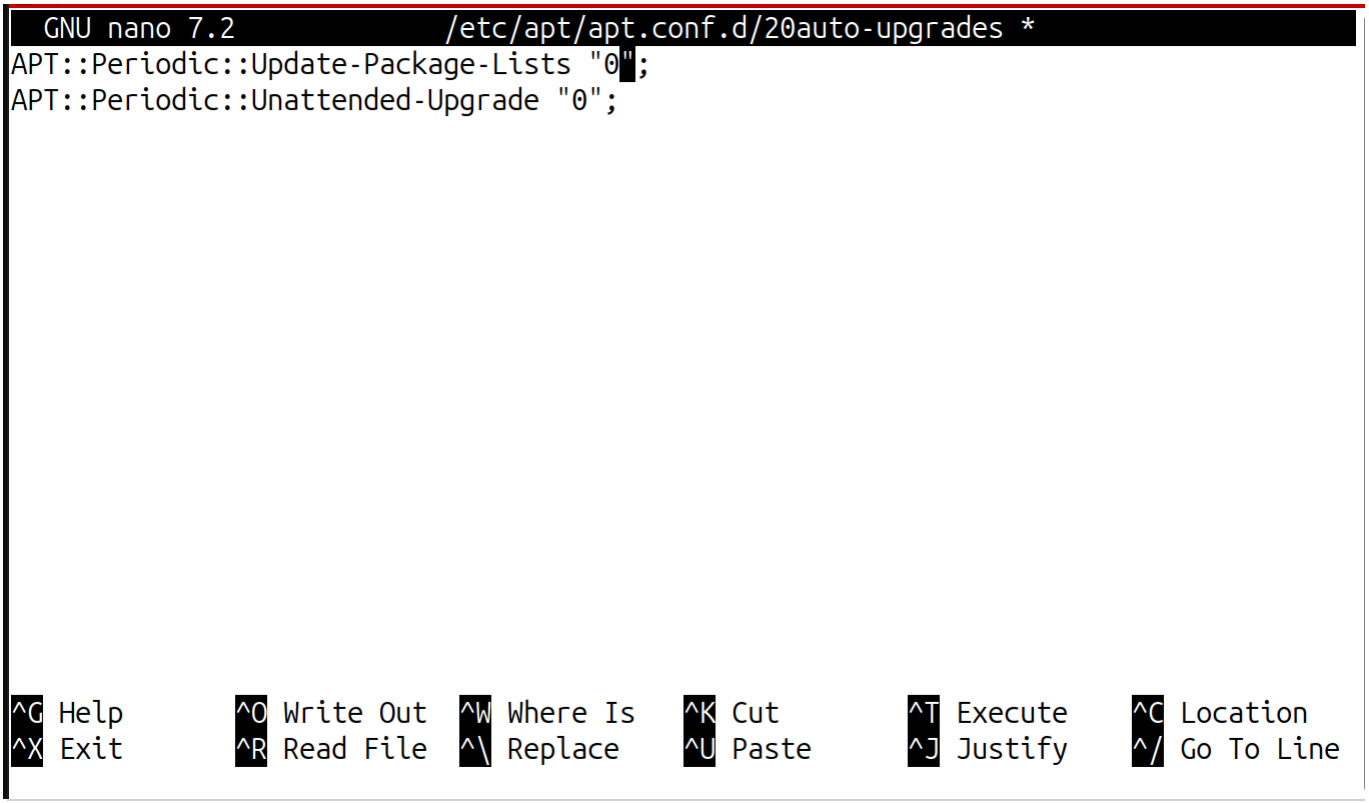
- yes 입력 후 SBC의 암호 입력

```
mkh@mkh:~$ ssh ubuntu@192.168.0.12
The authenticity of host '192.168.0.12 (192.168.0.12)' can't be established.
ED25519 key fingerprint is: SHA256:wfQ7Ie6wuamIf7a7lpZNEcTv9f+kV6hhgVG4/XdrDAs
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.0.12' (ED25519) to the list of known hosts.
ubuntu@192.168.0.12's password:
```

[SBC] 우선 자동 업데이트 끄기

Code

```
1  sudo nano /etc/apt/apt.conf.d/20auto-upgrades
```



[SBC] 부팅 지연 방지 설정

네트워크 없는 경우에도 부팅 지연을 방지하도록 설정

Code

```
sudo systemctl mask systemd-networkd-wait-online.service
```

[SBC] 일시 중단 및 최대 절전 모드 사용 안 함으로 설정

Code

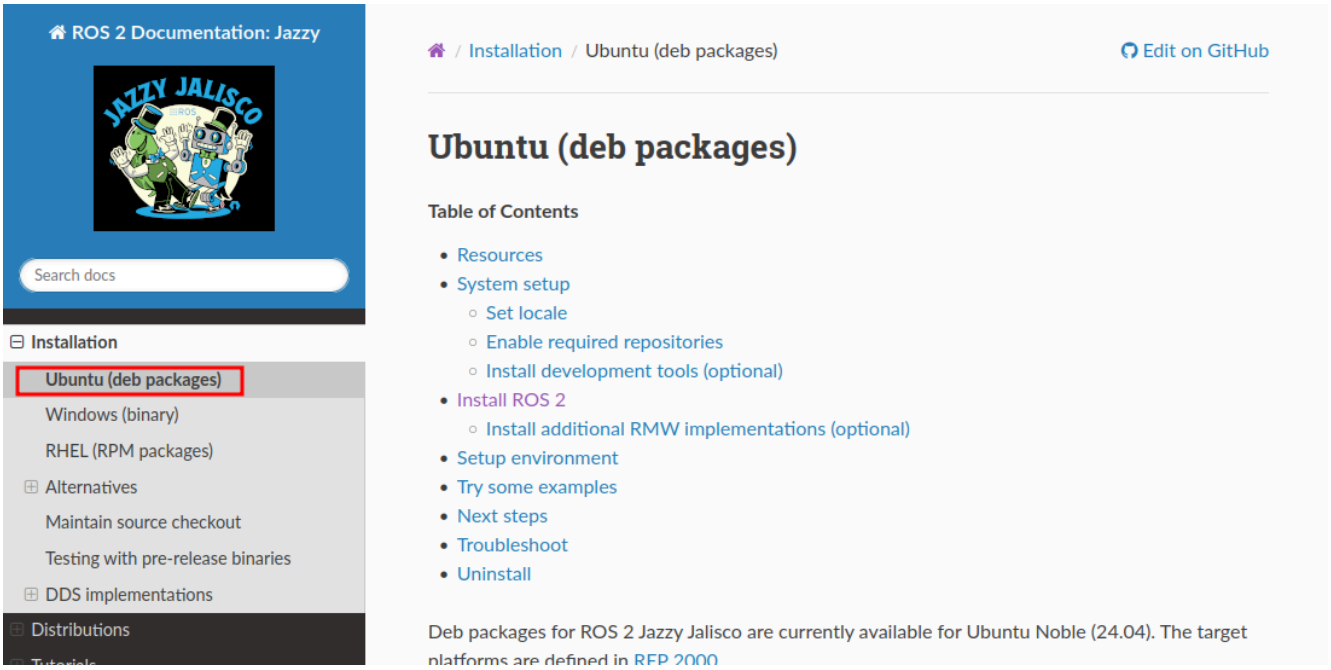
```
sudo systemctl mask sleep.target suspend.target hibernate.target hybrid-sleep.target
```

SBC에 ROS2 환경 구축하기

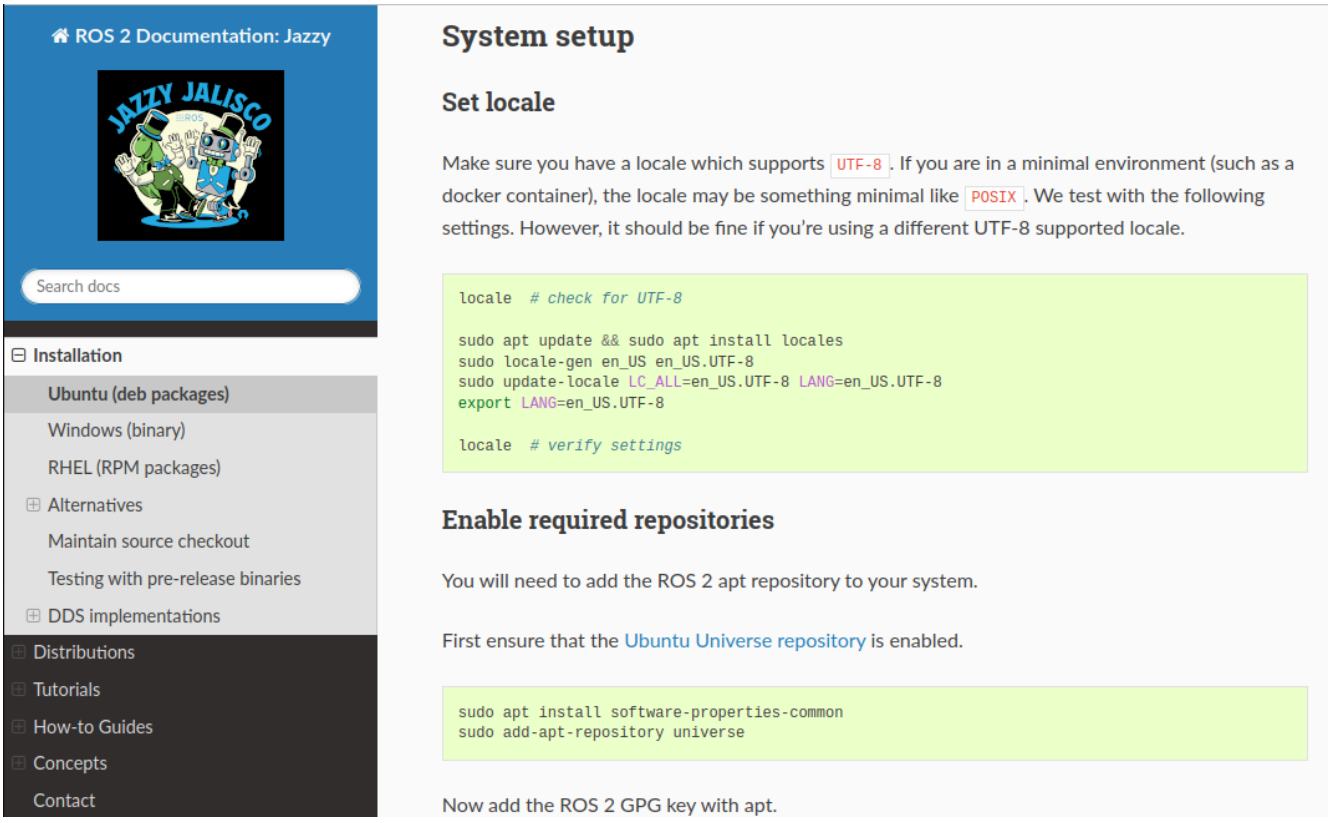
ROS2 Jazzy 설치 페이지

https://docs.ros.org/en/jazzy/index.html (https://docs.ros.org/en/jazzy/index.html)

- Ubuntu (deb packages) 선택



설치 과정 따라가기



[SBC] Set locale

- **locale**: 시스템의 언어, 문자 인코딩, 날짜/시간 형식 등을 정의하는 설정
- ROS2는 **UTF-8** 인코딩을 요구하므로 locale 설정이 필수
- Copy 해서 붙여넣기

Set locale

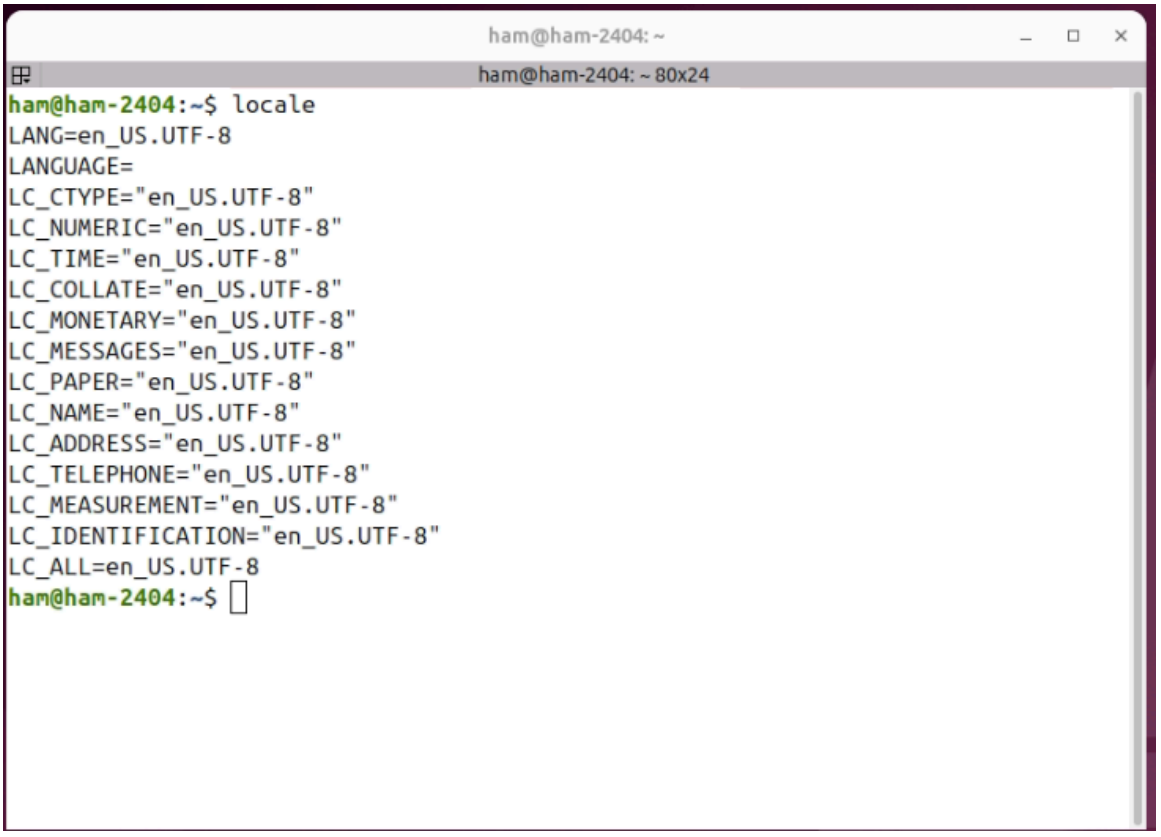
Make sure you have a locale which supports `UTF-8`. If you are in a minimal environment (such as a docker container), the locale may be something minimal like `POSIX`. We test with the following settings. However, it should be fine if you're using a different UTF-8 supported locale.

```
locale  # check for UTF-8

sudo apt update && sudo apt install locales
sudo locale-gen en_US en_US.UTF-8
sudo update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8
export LANG=en_US.UTF-8

locale  # verify settings
```

[SBC] locale 설정 확인



[SBC] universe 확인

- **universe**: Ubuntu의 커뮤니티 관리 오픈소스 패키지 저장소
- ROS2 패키지가 이 저장소를 통해 배포되므로 활성화 필요

```
sudo apt install software-properties-common
sudo add-apt-repository universe
```

[SBC] ROS 2 전용 APT 저장소를 자동으로 등록

```
$ sudo apt update && sudo apt install curl -y
$ export ROS_APT_SOURCE_VERSION=$(curl -s https://api.github.com/repos/ros-infrastructure/ros-apt-source/releases/latest | grep -F "tag_n
$ curl -L -o /tmp/ros2-apt-source.deb "https://github.com/ros-infrastructure/ros-apt-source/releases/download/${ROS_APT_SOURCE_VERSION}/r
$ sudo apt install /tmp/ros2-apt-source.deb
```

[SBC] apt update

새로 추가한 ROS2 저장소의 패키지 목록을 갱신합니다.

```
jt@ubuntu:~$ sudo apt update
Hit:1 https://packages.microsoft.com/repos/code stable InRelease
Hit:2 https://dl.google.com/linux/chrome/deb stable InRelease
Hit:3 http://kr.archive.ubuntu.com/ubuntu noble InRelease
Hit:4 http://kr.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:5 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:6 http://packages.ros.org/ros2/ubuntu noble InRelease [4,676 B]
Hit:7 http://kr.archive.ubuntu.com/ubuntu noble-backports InRelease
Get:8 http://packages.ros.org/ros2/ubuntu noble/main amd64 Packages [1,060 kB]
Fetched 1,064 kB in 3s (367 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
```

[SBC] Development tools 설치

- ROS2 패키지를 **소스에서 빌드**할 때 필요한 도구 모음
- **colcon**(빌드 도구), **rosdep**(의존성 관리), **vcstool**(버전 관리) 등 포함

```
jt@ubuntu:~$ sudo apt update && sudo apt install ros-dev-tools
Hit:1 https://packages.microsoft.com/repos/code stable InRelease
Hit:2 https://dl.google.com/linux/chrome/deb stable InRelease
Hit:3 http://packages.ros.org/ros2/ubuntu noble InRelease
Hit:4 http://kr.archive.ubuntu.com/ubuntu noble InRelease
Hit:5 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:6 http://kr.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:7 http://kr.archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

[SBC] apt upgrade 진행

Code

```
sudo apt upgrade
```

[SBC] ros-jazzy-ros-base 설치 시작

라즈베리파이에 서버용 Ubuntu OS를 설치했기 때문에 GUI 관련 툴이 필요 없다.

ROS-Base Install (Bare Bones): Communication libraries, message packages, command line tools. No GUI tools.

```
$ sudo apt install ros-jazzy-ros-base
```

SBC에 터틀봇 환경 구축하기

[SBC] 터틀봇 패키지의 의존성 패키지들 설치

Code

```
sudo apt install python3-argcomplete python3-colcon-common-extensions libboost-syste
m-dev build-essential
sudo apt install ros-jazzy-hls-lfcd-lds-driver ros-jazzy-turtlebot3-msgs ros-jazzy-d
ynamixel-sdk ros-jazzy-xacro
sudo apt install libudev-dev
```

[SBC] 터틀봇 패키지 설치

Code

```
mkdir -p ~/turtlebot3_ws/src && cd ~/turtlebot3_ws/src
git clone -b jazzy https://github.com/ROBOTIS-GIT/turtlebot3.git
git clone -b jazzy https://github.com/ROBOTIS-GIT/ld08_driver.git
git clone -b jazzy https://github.com/ROBOTIS-GIT/coin_d4_driver
cd ~/turtlebot3_ws/src/turtlebot3 && rm -r turtlebot3_cartographer turtlebot3_naviga
tion2
```

[SBC] 빌드

Code

```
cd ~/turtlebot3_ws/
echo 'source /opt/ros/jazzy/setup.bash' >> ~/.bashrc
source ~/.bashrc
colcon build --symlink-install --parallel-workers 1
```

[SBC] bashrc 에 추가

Code

```
echo 'source ~/turtlebot3_ws/install/setup.bash' >> ~/.bashrc
source ~/.bashrc
```

[SBC] OpenCR 연결용 udev 규칙 설정

Code

```
sudo cp `ros2 pkg prefix turtlebot3_bringup`/share/turtlebot3_bringup/script/99-turt
lebot3-cdc.rules /etc/udev/rules.d/
sudo udevadm control --reload-rules
sudo udevadm trigger
```

[SBC] ROS Domain ID 설정

- pc 와 같은 ROS_DOMAIN_ID 설정

Code

```
echo 'export ROS_DOMAIN_ID=30' >> ~/.bashrc
source ~/.bashrc
```

[SBC] LDS 모델 설정

보유한 LDS 모델에 맞게 환경변수를 설정

Code

```
# 사용 모델에 맞춰 하나만 설정
echo 'export LDS_MODEL=LDS-01' >> ~/.bashrc
# 또는
echo 'export LDS_MODEL=LDS-02' >> ~/.bashrc
# 또는
echo 'export LDS_MODEL=LDS-03' >> ~/.bashrc
source ~/.bashrc
```

참고

OpenCR 펌웨어 설정

[SBC] dependency 설치

Code

```
sudo dpkg --add-architecture armhf
sudo apt-get update
sudo apt-get install libc6:armhf
```

[SBC] 환경 변수 설정

가지고 있는 모델이 와플인 경우 burger 대신 waffle 로 변경

Code

```
echo "export TURTLEBOT3_MODEL=burger" >> ~/.bashrc
echo "export OPENCR_MODEL=burger" >> ~/.bashrc
echo "export OPENCR_PORT=/dev/ttyACM0" >> ~/.bashrc
source ~/.bashrc
```

[SBC] 펌웨어 다운로드 후 압축 풀기

Code

```
wget https://github.com/ROBOTIS-GIT/OpenCR-Binaries/raw/master/turtlebot3/ROS2/latest/opencr_update.tar.bz2
tar -xvf opencr_update.tar.bz2
```

[SBC] 펌웨어 업로드 하면 완료

Code

```
cd ./opencr_update
./update.sh $OPENCR_PORT $OPENCR_MODEL.opencr
```

```
ubuntu@ubuntu:~/opencr_update$ cd ./opencr_update
./update.sh $OPENCR_PORT $OPENCR_MODEL.opencr
-bash: cd: ./opencr_update: No such file or directory
aarch64
arm
OpenCR Update Start..
opencr_ld_shell ver 1.0.0
opencr_ld_main
[   ] file name          : burger.opencr
[   ] file size          : 136 KB
[   ] fw_name            : burger
[   ] fw_ver             : V230127R1
[OK] Open port           : /dev/ttyACM0
[   ]
[   ] Board Name         : OpenCR R1.0
[   ] Board Ver          : 0x17020800
[   ] Board Rev          : 0x00000000
[OK] flash_erase         : 1.12s
[OK] flash_write         : 1.35s
[OK] CRC Check           : D92222 D92222 , 0.004000 sec
[OK] Download
[OK] jump_to_fw
ubuntu@ubuntu:~/opencr_update$
```

터틀봇 동작 테스트

[SBC] 로봇에서 bringup 실행

Code

```
1  ros2 launch turtlebot3_bringup robot.launch.py
```

```
ubuntu@desktop:~$ ros2 launch turtlebot3_bringup robot.launch.py
[INFO] [launch]: All log files can be found below /home/ubuntu/.ros/log/20
24-08-06-15-38-14-170869-desktop-1899
[INFO] [launch]: Default logging verbosity is set to INFO
urdf_file_name : turtlebot3_waffle_pi.urdf
[INFO] [robot_state_publisher-1]: process started with pid [1900]
[INFO] [ld08_driver-2]: process started with pid [1902]
[INFO] [turtlebot3_ros-3]: process started with pid [1904]
[turtlebot3_ros-3] [INFO] [1722926295.300915205] [turtlebot3_node]: Init T
urtleBot3 Node Main
[turtlebot3_ros-3] [INFO] [1722926295.302007094] [turtlebot3_node]: Init D
ynamixelSDKWrapper
[turtlebot3_ros-3] [INFO] [1722926295.309993779] [DynamixelSDKWrapper]: Su
cceeded to open the port(/dev/ttyACM0)!
[turtlebot3_ros-3] [INFO] [1722926295.318406890] [DynamixelSDKWrapper]: Su
cceeded to change the baudrate!
[robot_state_publisher-1] [INFO] [1722926295.355558631] [robot_state_publi
sher]: got segment base_footprint
[robot_state_publisher-1] [INFO] [1722926295.355946112] [robot_state_publi
sher]: got segment base_link
[robot_state_publisher-1] [INFO] [1722926295.356014019] [robot_state_publi
sher]: got segment base_scan
[robot_state_publisher-1] [INFO] [1722926295.356049668] [robot_state_publi
sher]: got segment camera_link
```

[SBC] 아래와 같이 run 메시지가 출력되면 bringup 완료

```
[turtlebot3_ros-3] [INFO] [1723012399.729643973] [turtlebot3_node]: Calibration End
[turtlebot3_ros-3] [INFO] [1723012399.729768640] [turtlebot3_node]: Add Motors
[turtlebot3_ros-3] [INFO] [1723012399.730313087] [turtlebot3_node]: Add Wheels
[turtlebot3_ros-3] [INFO] [1723012399.730806015] [turtlebot3_node]: Add Sensors
[turtlebot3_ros-3] [INFO] [1723012399.737617624] [turtlebot3_node]: Succeeded to create battery state publisher
[turtlebot3_ros-3] [INFO] [1723012399.742116776] [turtlebot3_node]: Succeeded to create imu publisher
[turtlebot3_ros-3] [INFO] [1723012399.746027593] [turtlebot3_node]: Succeeded to create sensor state publisher
[turtlebot3_ros-3] [INFO] [1723012399.748445475] [turtlebot3_node]: Succeeded to create joint state publisher
[turtlebot3_ros-3] [INFO] [1723012399.748567494] [turtlebot3_node]: Add Devices
[turtlebot3_ros-3] [INFO] [1723012399.748610032] [turtlebot3_node]: Succeeded to create motor power server
[turtlebot3_ros-3] [INFO] [1723012399.751544528] [turtlebot3_node]: Succeeded to create reset server
[turtlebot3_ros-3] [INFO] [1723012399.754058818] [turtlebot3_node]: Succeeded to create sound server
[turtlebot3_ros-3] [INFO] [1723012399.756624591] [turtlebot3_node]: Run!
[turtlebot3_ros-3] [INFO] [1723012399.797731799] [diff_drive_controller]: Init Odometry
[turtlebot3_ros-3] [INFO] [1723012399.811079440] [diff_drive_controller]: Run!
```

[PC] PC에서 터틀봇 teleop 노드 실행

w a s d x 키입력으로 속도값을 설정해서 주행 테스트

Code

```
ros2 run turtlebot3_teleop teleop_keyboard
```

```
mkh@mkh:~$ ros2 run turtlebot3_teleop teleop_keyboard

Control Your TurtleBot3!
---------------------------
Moving around:
        w
   a    s    d
        x

w/x : increase/decrease linear velocity (Burger : ~ 0.22, Waffle and Waffl
e Pi : ~ 0.26)
a/d : increase/decrease angular velocity (Burger : ~ 2.84, Waffle and Waff
le Pi : ~ 1.82)

space key, s : force stop

CTRL-C to quit

currently:      linear velocity 0.01     angular velocity 0.0
```

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